SOC Detection & Response Automation Lab

Project Report

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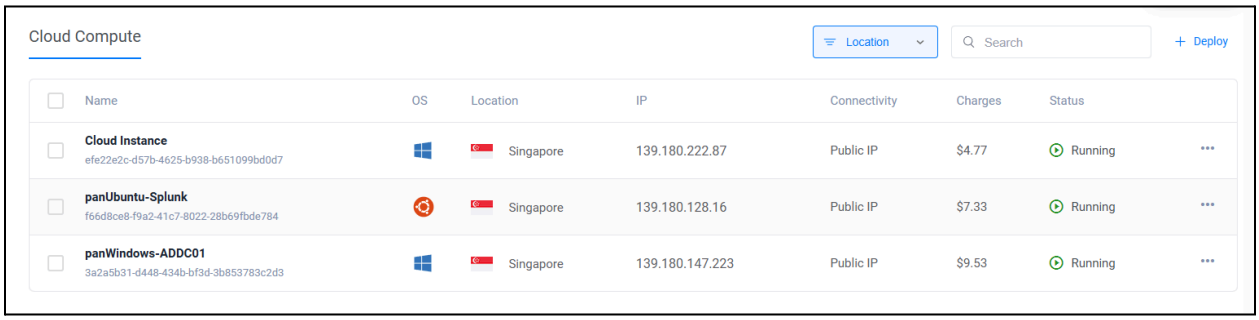
Project Type: Cybersecurity / SOC Lab
Technologies: Splunk SIEM, Shuffle SOAR, Active Directory, Windows Server, Slack
Focus Area: Security Monitoring, Threat Detection, Automated Incident Response

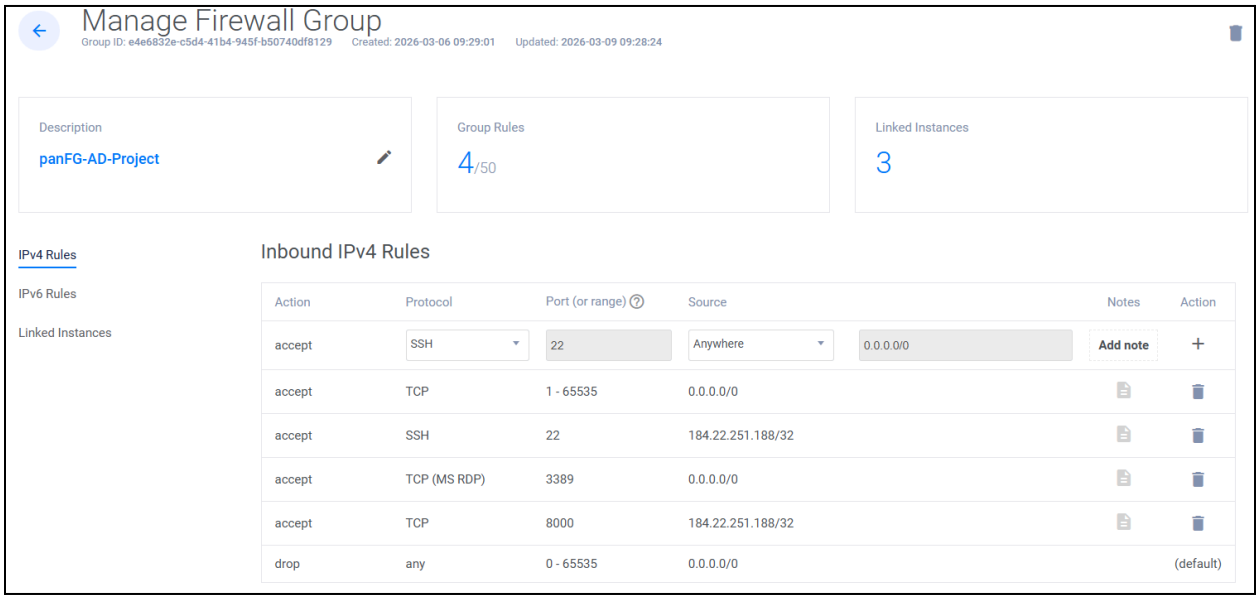
1. Project Overview

This project demonstrates a Security Operations Center (SOC) monitoring and automated incident response workflow designed to detect and respond to suspicious authentication activity within a Windows Active Directory environment.

The system collects Windows authentication logs and analyzes them using a Security Information and Event Management (SIEM) platform. When suspicious Remote Desktop Protocol (RDP) login activity is detected, the system automatically triggers a response workflow that notifies a SOC analyst and allows them to take remediation actions.

The objective of this project is to simulate a real-world SOC environment where security alerts are detected, investigated, and mitigated through automation and analyst interaction.





2. Lab Architecture

The lab environment consists of several components designed to simulate a corporate network monitored by a SOC.

Infrastructure Components

Attacker Machine: Simulates an external system attempting to authenticate to the internal network.

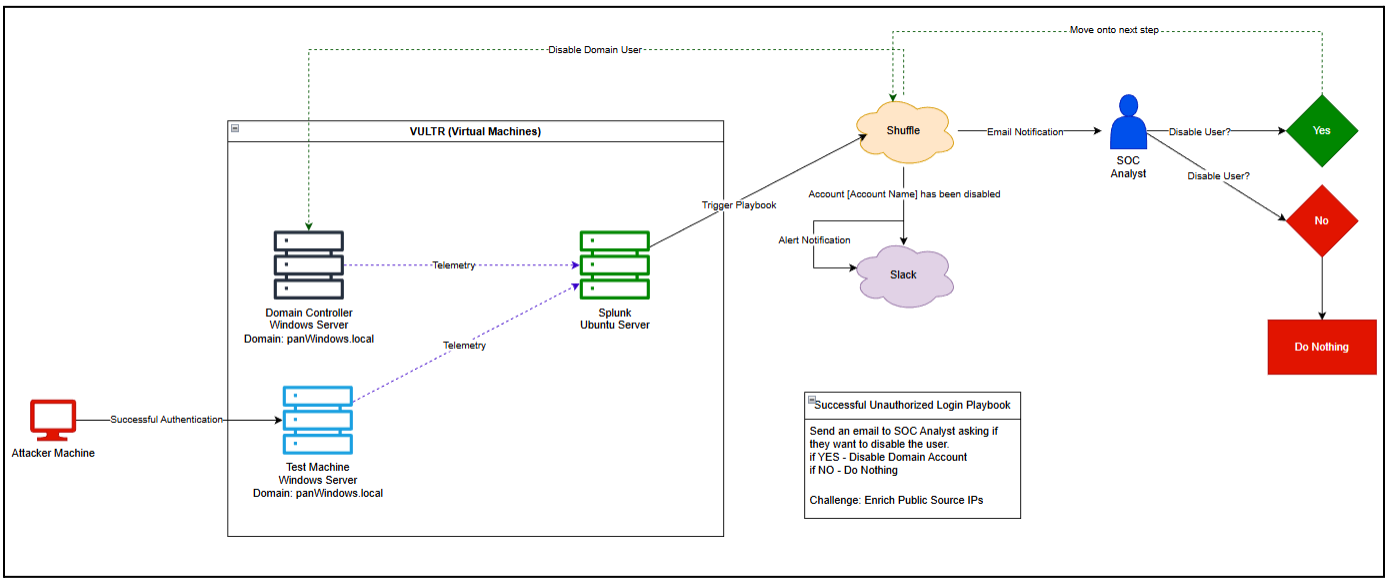
Domain Controller: Windows Server acting as the Active Directory Domain Controller. Responsible for authentication and domain user management.

Test Machine: Windows Server joined to the Active Directory domain. Used to simulate login activity and RDP sessions.

Splunk Server: Ubuntu-based server used for centralized log collection and analysis. Receives Windows Security Event logs and performs security monitoring.

Shuffle SOAR Platform: Security orchestration platform used to automate incident response. Executes playbooks triggered by security alerts.

Slack Notification System: Used to notify SOC analysts of security alerts and allow interaction with automated workflows.



3. Log Collection and Monitoring

The project collects Windows Security Event logs generated by authentication activity within the domain environment.

Log Sources

Windows servers generate authentication events which include important information such as:

- Username
- Source IP address
- Authentication type
- Login success or failure
- Timestamp of the event

These logs are forwarded to the SIEM platform where they are indexed and analyzed.

Key Event Types

The primary events analyzed in this project include:

Event ID 4624 – Successful Logon
Indicates that a user successfully authenticated to the system.

Logon Type 10 – Remote Interactive Logon
Represents login via Remote Desktop Protocol (RDP).

These events are useful for detecting suspicious remote login activity originating from external sources.

4. Detection Logic

The SIEM platform is configured with detection queries that identify suspicious successful RDP login events.

Detection Query Example

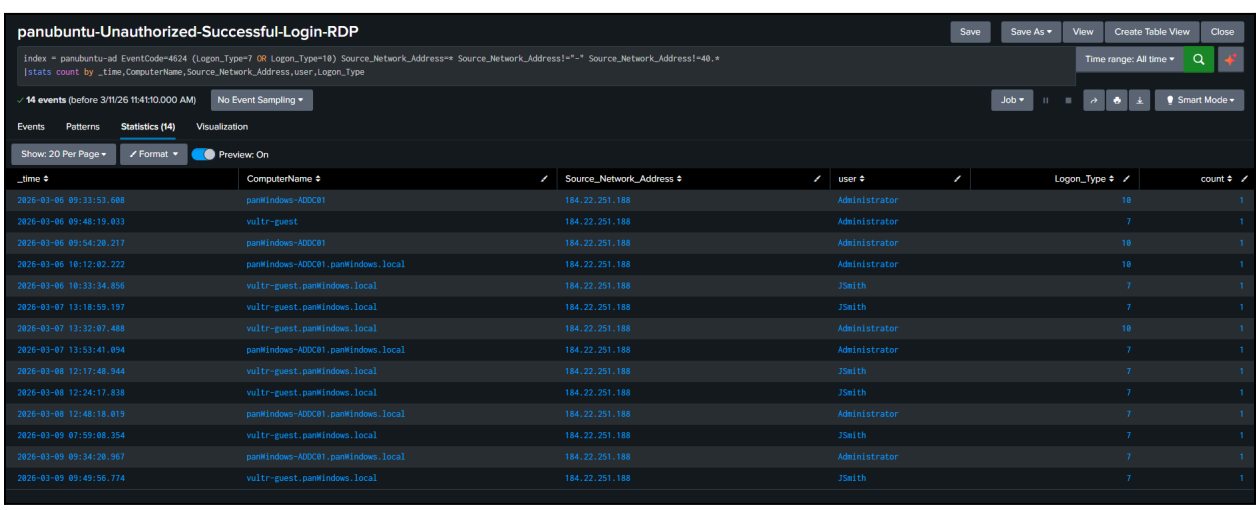
```
index = panubuntu-ad EventCode=4624 (Logon_Type=7 OR Logon_Type=10)
Source_Network_Address=*
Source_Network_Address!="-"
Source_Network_Address!=40.*
 | stats count by _time,ComputerName,Source_Network_Address,user,Logon_Type
```

Detection Goals

The query is designed to:

- Identify successful authentication events
- Detect Remote Desktop logins
- Extract source network addresses
- Highlight login activity originating from external IP addresses

This allows analysts to quickly identify potential unauthorized remote access attempts.



_time	ComputerName	Source_Network_Address	user	Logon_Type	count
2026-03-06 09:33:53.608	panWindows-ADDC01	184.22.251.188	Administrator	10	1
2026-03-06 09:48:19.033	vultr-guest	184.22.251.188	Administrator	7	1
2026-03-06 03:54:20.217	panWindows-ADDC01	184.22.251.188	Administrator	10	1
2026-03-06 10:12:02.222	panWindows-ADDC01.panWindows.local	184.22.251.188	Administrator	10	1
2026-03-06 10:33:34.856	vultr-guest.panWindows.local	184.22.251.188	JSmith	7	1
2026-03-07 13:18:59.197	vultr-guest.panWindows.local	184.22.251.188	JSmith	7	1
2026-03-07 13:32:07.488	vultr-guest.panWindows.local	184.22.251.188	Administrator	10	1
2026-03-07 13:53:41.094	panWindows-ADDC01.panWindows.local	184.22.251.188	Administrator	7	1
2026-03-08 12:17:48.944	vultr-guest.panWindows.local	184.22.251.188	JSmith	7	1
2026-03-08 12:24:17.838	vultr-guest.panWindows.local	184.22.251.188	JSmith	7	1
2026-03-08 12:48:18.019	panWindows-ADDC01.panWindows.local	184.22.251.188	Administrator	7	1
2026-03-09 07:59:08.354	vultr-guest.panWindows.local	184.22.251.188	JSmith	7	1
2026-03-09 09:34:20.967	panWindows-ADDC01.panWindows.local	184.22.251.188	Administrator	7	1
2026-03-09 09:49:56.774	vultr-guest.panWindows.local	184.22.251.188	JSmith	7	1

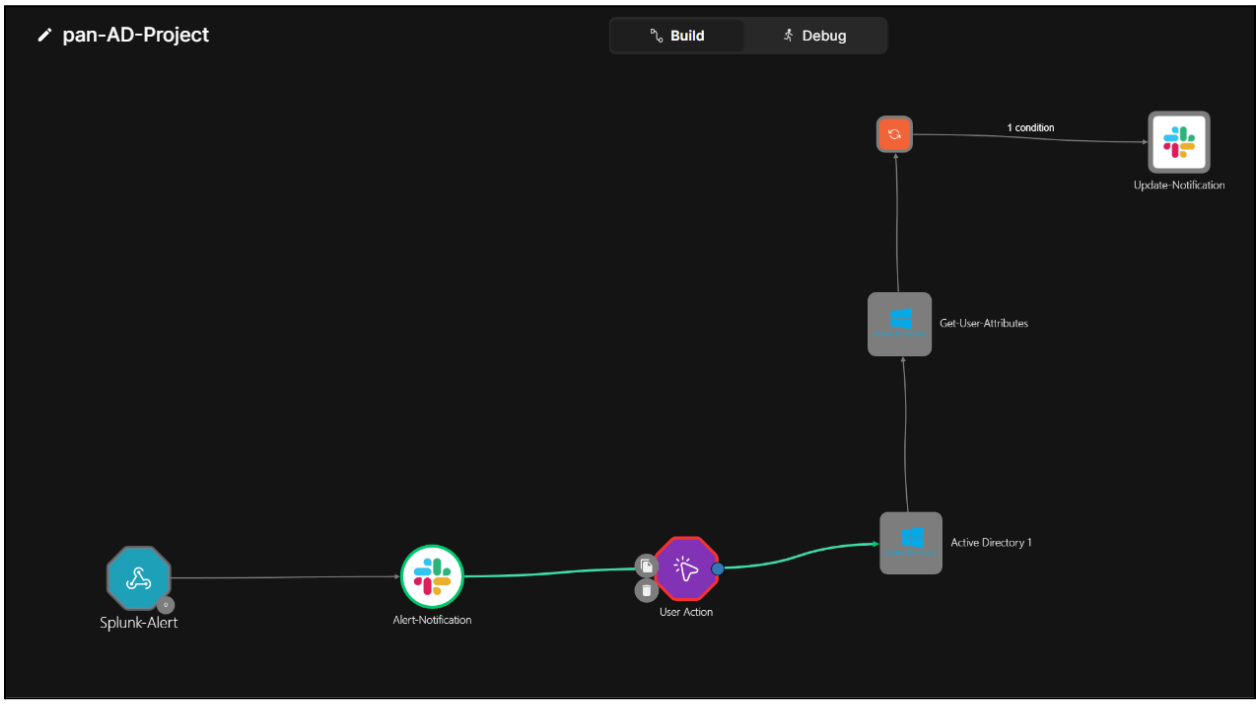
5. SOAR Automation Workflow

Once a detection rule identifies suspicious activity, the SIEM system triggers an automated workflow through the SOAR platform.

Playbook Workflow

1. **Alert Trigger**
   - The SIEM generates an alert when a suspicious login is detected.
2. **Slack Notification**
   - A message is sent to the SOC analyst via Slack containing information about the login event.
3. **User Investigation**
   - The workflow retrieves user information from Active Directory.
4. **Analyst Decision**
   - The SOC analyst reviews the alert and decides whether the account should be disabled.
5. **Automated Response**
   - If confirmed malicious, the playbook disables the affected domain user account.

This process demonstrates how automation can accelerate the incident response process while maintaining human oversight.



6. Incident Response Process

The project simulates a simplified SOC incident response workflow.

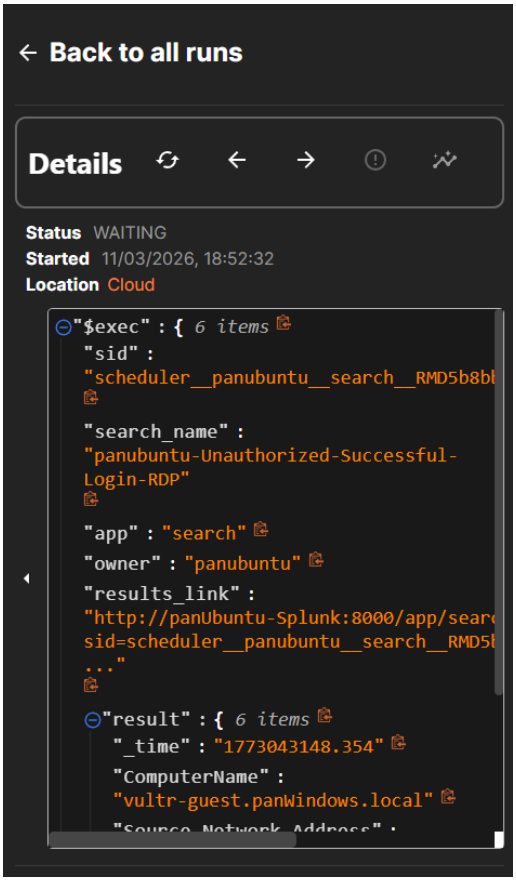
Step 1 – Detection
The SIEM detects suspicious authentication events based on predefined rules.

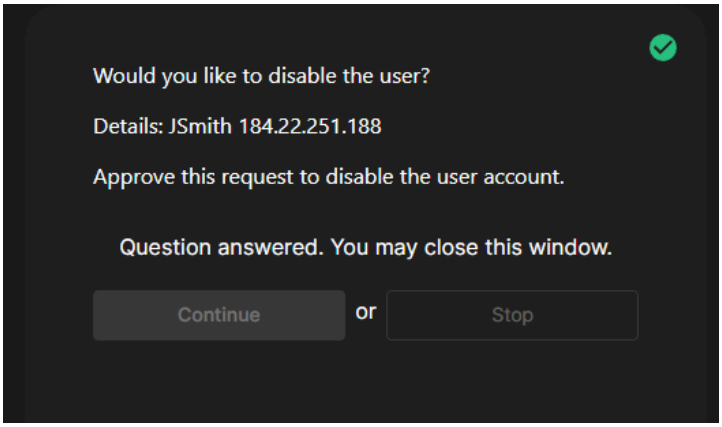
Step 2 – Alerting
An alert is automatically generated and forwarded to the SOC team.

Step 3 – Investigation
The SOC analyst reviews contextual information such as:

- Source IP address
- Target system
- Username involved
- Login method

Step 4 – Response
If the activity is determined to be malicious, the compromised user account is disabled via the automated playbook.





Shuffle Support <shuffle-support@shuffler.io>
ถึง ฉัน

18:52 (10 นาทีที่ผ่านมา)

Action required!
Would you like to disable the user?

Details:
JSmith
184.22.251.188

Approve this request to disable the user account.

If this is TRUE click this: https://shuffler.io/workflows/f1d19a7e-eafe-4c5c-a02b-60836662a02e/run?authorization=d0a41b49-0e23-4dac-9201-ab60621c9513&start=36864098-b0e1-45e2-8166-b3370639f9ab&reference_execution=6009550c-35d8-4ba7-80b3-a4ba97d860ad&answer=true

IF THIS IS FALSE, click this: https://shuffler.io/workflows/f1d19a7e-eafe-4c5c-a02b-60836662a02e/run?authorization=d0a41b49-0e23-4dac-9201-ab60621c9513&start=36864098-b0e1-45e2-8166-b3370639f9ab&reference_execution=6009550c-35d8-4ba7-80b3-a4ba97d860ad&answer=false

...

pan-AD-Project

Get 50% off Business+
4 days left on this offer

Huddles
Directories

Starred
Drag and drop important stuff here

Channels
alerts
all-pan-ad-project
new-alerts
new-channel
social
+ Add channels

Direct messages
p51119.2013 you

 # new-alerts

Messages | Add canvas +

User: JSmith
Source IP: 184.22.251.188

Alert: panubuntu-Unauthorized-Successful-Login-RDP
Time: 1773043148.354
User: JSmith
Source IP: 184.22.251.188

Shuffle-SOAR-Alert APP 6:43 PM
Alert: panubuntu-Unauthorized-Successful-Login-RDP
Time: 1773043148.354
User: JSmith
Source IP: 184.22.251.188

Shuffle-SOAR-Alert APP 6:52 PM
Alert: panubuntu-Unauthorized-Successful-Login-RDP
Time: 1773043148.354
User: JSmith
Source IP: 184.22.251.188

Account: JSmith has been disabled

Shuffle-SOAR-Alert APP 6:52 PM
Alert: panubuntu-Unauthorized-Successful-Login-RDP
Time: 1773043148.354
User: JSmith
Source IP: 184.22.251.188

Account: JSmith has been disabled

Guest	User	Built-in account for gue...
Jenny Smith	User	

7. Results

The system successfully demonstrates an end-to-end SOC monitoring pipeline that includes:

- Security log collection from Windows servers
- Detection of suspicious authentication activity
- Automated alert generation
- Analyst notification via Slack
- Automated response through account containment

The project highlights how SIEM and SOAR technologies can work together to improve security monitoring and incident response capabilities.

8. Conclusion

This project demonstrates the practical implementation of a SOC detection and response workflow using SIEM and SOAR technologies.

By combining centralized log monitoring, automated alerting, and response playbooks, the system is capable of detecting suspicious login behavior and taking rapid containment actions.

The project reflects common practices used by real-world SOC teams and highlights the importance of automation in modern cybersecurity operations.